

Matthew McCall

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Skills: C++, CMake, CMSIS, FreeRTOS, GNU/Linux, Git, MATLAB, KiCad, SPICE, SolidWorks, $\LaTeX$

Education

Rensselaer Polytechnic Institute • Troy, NY

Bachelor of Science in Computer Science and Computer Systems Engineering

May 2026

- Completed the Computer Systems Design ECSE Focus Area.

Relevant Coursework: Space Vehicle Design, Microprocessor Systems, Robotics II

Work Experience

BETA Technologies • South Burlington, VT

June 2026 – Present

Software Engineer

- Working on airborne software.

Health Innovations Incubator & Technology Center • Watervliet, NY

March 2026 – May 2026

Part-Time Software Engineer

- Created a device for hospitalized children with disabilities to remotely express non-urgent needs to nurses.
- Implemented a scalable wireless network to support hundreds of devices in a hospital environment.

HarcoSemco • Branford, CT

May 2025 – August 2025

Engineering and Quality Intern

- Led the end-to-end development of Bluetooth-based tracking system, selected hardware, and developed C++, Java, and Kotlin software to monitor assets across a 12,000 sq. ft. facility with 8-foot positional accuracy.

Scientific Computation Research Center • Troy, NY

August 2022 – May 2026

Research Assistant

- Implemented a GPU memory pool to reduce runtime by 30% for Omega_h, a C++ library for mesh adaptivity.
- Developed an algorithm to find the nearest simplex to a point for PCMS, a C++ mesh coupling library.
- Authored a scientific paper in $\LaTeX$ and presented at the Spring 2024 RPI Undergraduate Research Symposium.

Mercer X Lab • Troy, NY

January 2025 – May 2025

Lab Assistant

- Mentored students in the use of oscilloscopes to analyze circuits, in soldering, and in lab safety best practices.
-

Project Experience

Open Algebra Software for Inferring Solutions (OASIS)

- Lead the development of the OASIS project, an open-source C++ library for computer algebra.
- Facilitated an AGILE-based Kanban workflow using GitHub Issues, Projects, and Pull Requests.
- Deployed GitHub Actions CI/CD pipelines to enforce unit and memory tests pass before PRs are merged.

AIN'T Is Not TRON

- AINT is a real-time, multiplayer game leveraging C & C++ to demonstrate CMSIS, FreeRTOS, UART, and interrupt-driven programming on the STM32F7. AINT was selected to present at an admissions event.

Helpful Assessment Readiness Platform

- Won Best AI Hack and Best Use of DigitalOcean Gradient AI at HackRPI 2025 by developing a mock interview platform that uses AI agents to simulate interviewer personas and provide automated feedback to users.

Eye Robot

- Eye Robot is a robot to assist visually impaired users. Collaborated on a team to develop C# and C++ software to perform mesh reconstruction from LiDAR and transmit navigation commands to the onboard ESP32.